

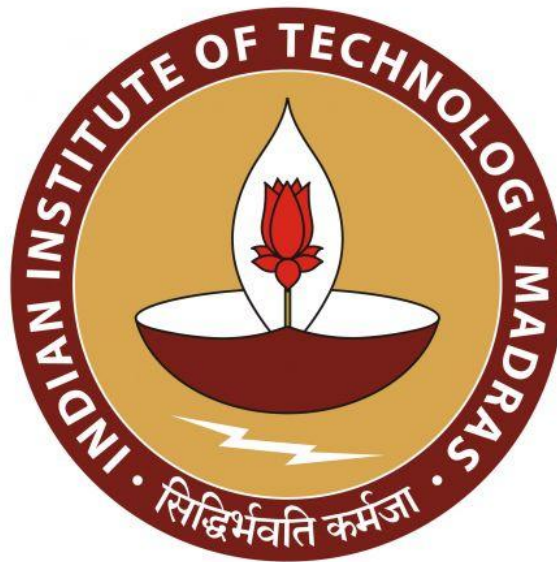
Improving Marketing Efficiency Through Customer Segmentation for a UK Online Retailer

A Proposal report for the BDM capstone Project

Submitted by

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Declaration Statement

I am working on a Project Title “**Improving Marketing Efficiency Through Customer Segmentation for a UK Online Retailer**”. I extend my appreciation to the **UCI Machine Learning Repository** for providing the publicly available secondary dataset that enabled me to conduct my project.

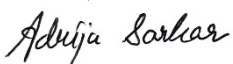
I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from **secondary sources** and has been carefully checked, cleaned, and analysed to ensure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilised for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Signature of Candidate: 

Name: ADRIJA SARKAR

Date: 13/11/2025

1. Executive Summary

This project focuses on a UK-based and registered non-store online retail store that mainly sells unique all-occasion gifts. The clientele consists primarily of wholesalers as well as individual consumers; hence, the business can be described as being active in both B2B and B2C sectors.

The organisation operates in a highly competitive online marketplace and handles a large volume of transactions. This makes it essential for the business to understand customer behaviour for effective marketing decision-making. To the contrary, the data obtained from the UCI Machine Learning Repository, consisting of the business's transactional records from 01/12/2010 to 09/12/2011, reveals the absence of a structured customer segmentation system. As a result, a key challenge faced by the business is the uniform distribution of marketing resources across all customers, which leads to inefficient spending and limited impact on revenue growth.

This project seeks to address this issue by identifying meaningful customer segments. Excel will be used for initial data cleaning and descriptive summaries, while Python will be used to compute Recency, Frequency, and Monetary (RFM) metrics and perform K-Means clustering. These methods will help uncover groups such as loyal high-value customers, infrequent buyers, and at-risk segments.

The expected outcomes will help the business improve market efficiency through data-driven customer targeting strategies and actionable insights that support better allocation of promotional resources, ultimately contributing to higher customer retention and profitability.

2. Organisation Background

This project examines a UK-registered, non-store online retailer specialising in unique gifts. Information regarding the organisation, established in 1981, was sourced from the [UCI Machine Learning Repository](#) and an associated academic article named [“Data mining for the online retail industry: A case study of RFM model-based customer segmentation using data mining”](#).

Initially operating via direct mail catalogues and telephone orders, the company (employing approx. 80 staff) launched its e-commerce platform around 2010. It now serves individual and wholesale clients across the UK and Europe, utilising its own website and Amazon.co.uk.

The dataset used in this project contains transactional records collected between 01/12/2010 and 09/12/2011, consisting of 541,909 instances and 8 variables.

3. Problem Statement

3.1 Lack of targeted customer segmentation:

The online retailer currently treats all customers uniformly, without distinguishing between high-value, frequent purchasers and low-engagement buyers, leading to inefficient utilisation of marketing resources.

3.2 Inefficient allocation of marketing expenditure:

Because customer behaviour has not been analysed systematically, promotional efforts and marketing investments are distributed evenly across all customers, reducing potential returns and limiting revenue growth.

3.3 Absence of a data-driven customer management strategy:

The organisation does not employ a structured analytical approach to evaluate customer purchase patterns, spending levels, or recency of transactions, restricting its ability to prioritise key customer groups and design effective retention strategies.

4. Background of the Problem

Primarily, the retailer lacks targeted customer segmentation because, despite possessing substantial transactional data, it fails to analyse purchasing patterns to identify high-value, frequent, or at-risk customers. This internal knowledge gap contrasts with external industry expectations for personalisation, where competitors utilise data-driven segmentation.

Consequently, marketing expenditure is allocated inefficiently. Internal operations rely on uniform campaign deployment, lacking an understanding of customer value, which wastes resources on non-responsive segments. This is untenable externally, as rising digital marketing costs and sophisticated competitor targeting render generic outreach ineffective.

Finally, the absence of a data-driven management strategy is due to an internal failure to utilise available, detailed variables (such as InvoiceNo, Quantity, CustomerID, and Country) to assess recency, loyalty, or spending potential. This internal gap conflicts with external e-commerce trends that demand retention, personalisation, and structured behavioural analysis.

5. Problem-Solving Approach

5.1. Tools for Analysis:

5.1.1. Microsoft Excel:

- 5.1.1.1. Perform initial data cleaning (remove cancellations, null CustomerIDs, and formatting issues).
- 5.1.1.2. Generate descriptive summaries using pivot tables to observe basic purchasing patterns.
- 5.1.1.3. Conduct preliminary frequency and monetary analysis to validate data trends.

5.1.2. Python (Pandas, NumPy, Scikit-learn, Matplotlib):

- 5.1.2.1. Transform invoice-level data into customer-level datasets.
- 5.1.2.2. Compute RFM (Recency, Frequency, Monetary) metrics per CustomerID.
- 5.1.2.3. Standardise variables and perform clustering using machine learning methods.
- 5.1.2.4. Visualise customer segments for comparison and profiling.

5.2. Solution Approach for Each Problem Statement:

5.2.1. Addressing “Lack of targeted customer segmentation”:

- 5.2.1.1. Use Python to aggregate all invoices for each customer.
- 5.2.1.2. Calculate **Recency** (days since last purchase), **Frequency** (number of invoices), and **Monetary Value** (total customer spend).
- 5.2.1.3. Standardise RFM values to ensure comparability across customers creating a structured numerical foundation for segmentation.

5.2.2. Addressing “Inefficient allocation of marketing expenditure”:

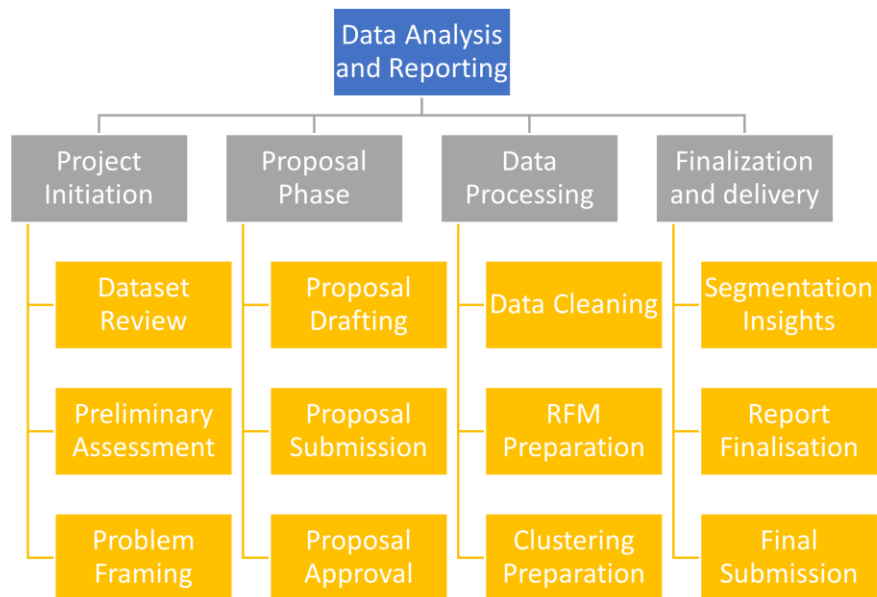
- 5.2.2.1. Apply **K-Means clustering** on standardised RFM metrics to identify meaningful customer groups.
- 5.2.2.2. Interpret clusters to differentiate: **High-value frequent purchasers**, **New or occasional buyers**, **Low-engagement or at-risk customers**.
- 5.2.2.3. Compare spending patterns and recency gaps across clusters to determine where marketing investment yields high or low returns.

5.2.3. Addressing “Absence of a data-driven customer management strategy”:

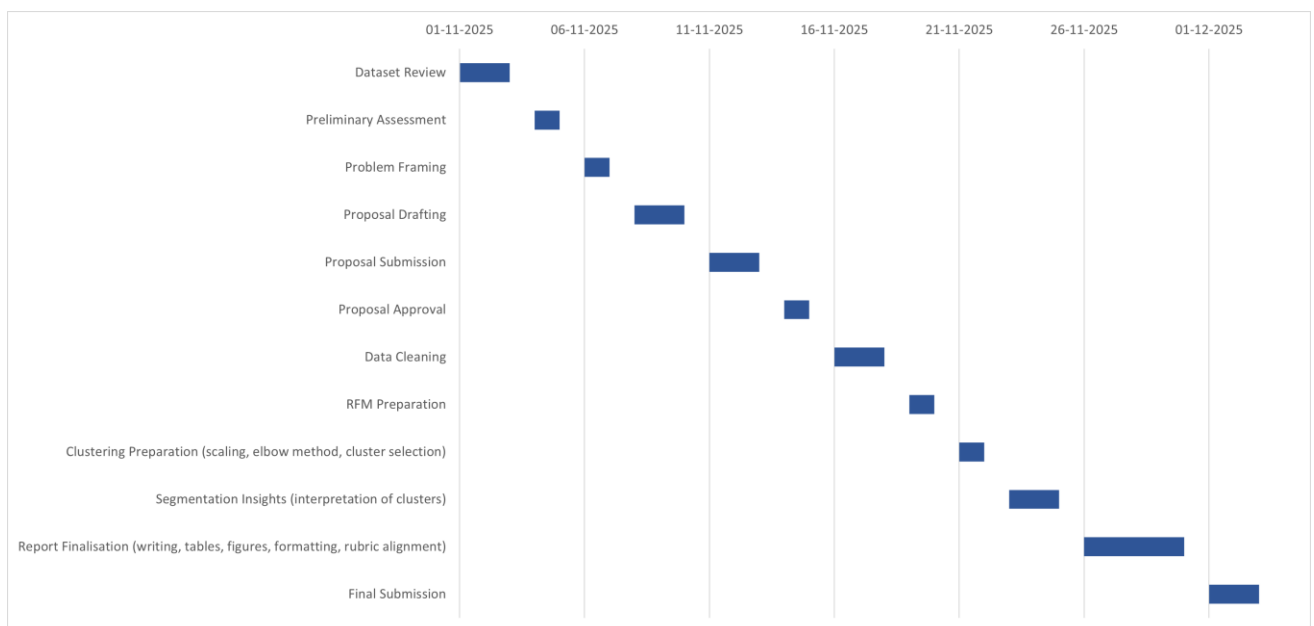
- 5.2.3.1. Analyse segment-level behaviours (average monetary value, purchasing intervals, transaction frequency).
- 5.2.3.2. Use Excel and Python visualisations to highlight behavioural differences.
- 5.2.3.3. Develop data-driven marketing actions for each segment, such as retention campaigns for high-value customers, reactivation offers for at-risk groups, and low-cost outreach for low-frequency buyers.

6. Expected Timeline

6.1. Work Breakdown Structure:



6.2. Gantt Chart:



Note. Expected timeline for completion of project.

7. Expected Outcome

7.1. Identification of distinct customer segments

Enables the organisation to clearly differentiate high-value loyal customers, frequent mid-value buyers, occasional shoppers, and at-risk customers based on RFM and clustering analysis.

7.2. Improved marketing resource allocation

Supports more efficient deployment of marketing budgets by targeting high-impact customer groups with suitable promotional strategies rather than investing uniformly across all customers.

7.3. Enhanced customer retention and engagement strategies

Provides insights to design personalised communication, loyalty initiatives, and reactivation campaigns tailored to each customer segment's behavioural patterns.

7.4. Data-driven decision-making for long-term growth

Empowers the business to transition from intuition-based marketing to evidence-based customer management, ultimately improving marketing ROI, boosting engagement, and supporting sustainable revenue growth.